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Full Length Article

Clustering of firms based on environmental, social, and governance ratings: Evidence from BIST sustainability index

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Abstract

In this paper, companies listed on the Borsa Istanbul (BIST) Sustainability Index are analyzed by performing a cluster analysis based on their environmental, social, and governance (ESG) scores. The results prove that firms with higher ESG ratings do not necessarily perform well in all ESG aspects. The outcomes of the cluster analysis reveal that firms with higher environmental and social scores are the cluster with the most prominent firms in terms of size but with low profitability. However, the group that scored poorly in environmental and social practices but the highest governance pillar was the highest performing in terms of the return on assets. This paper highlights the significance of forming clusters and linking sustainability practices with performance characteristics.
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Keywords: BIST Sustainability index; Cluster analysis; Environmental; Social; And governance (ESG) ratings

1. Introduction

In the past two decades, environmental degradation and climate change increased, which affected public views about investment and compelled firms to reflect these changes in their activities (Azmi et al., 2019). Accordingly, firms were obliged to abide by environmental regulations and report their strategies for addressing environmental degradation. As the twenty-first century began, various corporate scandals emerged regarding fake disclosures and fraud, causing huge losses for their investors. This change resulted in a loss of confidence and caused hesitation about equity investment. Consequently, firms had to reassure investors about the strength of their corporate governance structures and report information related to these mechanisms, which promise excellent management by firms to protect stakeholder rights. These developments led to the calculation of Environmental, Social Governance (ESG) Index scores in almost all equity markets worldwide. Moreover, the COVID-19 pandemic stressed the interconnectedness between sustainability, climate change, corporate profits, and human health, highlighting the role of ESG investing (Adams & Abhayawansa, 2022). Companies display their sustainability indexes as evidence of their ESG practices and participate in these ESG indexes to attract individual and institutional investors (Scalet & Kelly, 2010).

Despite the contraction in funding during the COVID-19 pandemic, over $500 billion was invested in funds with ESG criteria in 2021, causing growth of 55 percent in ESG-oriented products (Morningstar, 2021). ESG investments and firms with higher ESG ratings had lower risk and were more resilient during periods of economic instability (Ferriani & Natoli, 2020). Some papers have concluded that these investments provide higher returns and have lower volatility than their traditional counterparts (Cunha et al., 2020; Diaz et al., 2021; Verheyden et al., 2016).

ESG investment (sustainable or socially responsible investment) is an investment strategy for investors who consider firms' attitudes toward environmental, social, and corporate

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governance factors, such as policies on climate change and human rights in seeking to earn higher returns and implement portfolio allocation strategies. ESG investment has gained attention from investors and policy makers and has shown significant growth in the past few decades. The increase in sustainable investment indicates that the trend toward ESG portfolios has risen despite the pandemic. This trend suggests that ESG portfolios are considered safe instruments even in periods of turmoil. The pandemic triggered interest by investors in the ESG aspects of companies during this period of uncertainty (Pastor & Vorsatz, 2020).

Sustainability indexes play a significant role in guiding investors about their ESG aspects (Arribas et al., 2021). ESG indexes consider the sustainability approaches of the firms in terms of their three pillars: environmental, social, and governance. The environmental pillar considers the attitudes and practices of the firms toward environmental issues such as climate change, air and water pollution, deforestation, and biodiversity. Thus, this part of the index grades firm activities related to energy consumption, energy efficiency, carbon and hazardous gas emissions, waste, water, and resource management. The social pillar is related to the firms' policies associated with human rights protection, workplace and product safety, labor standards, gender policies, and public health issues. The governance pillar of the index concerns factors related to the good governance of firms, such as board independence, protection of shareholder rights, control and monitoring activities, anticompetitive procedures, and compliance with laws and regulations. Many studies associate ESG ratings with the firms' responses to environmental and social aspects such as corporate social performance (Drempetic et al., 2020).

ESG rating agencies evaluate the ESG performance of firms by demanding data on hundreds of criteria. Despite the popularity of the ESG ratings in evaluating sustainability performance, different rating agencies give conflicting ratings. Some firms indexed in ESG indexes are involved in scandals, which demonstrate unsustainable practices (Arribas et al., 2021). Arribas et al. (2019) highlight many irresponsible behaviors by firms listed in the Dow Jones Sustainability Index. Even though the overall ESG Ratings are considered for sustainability practices, it is clear that firms follow environmental, social, and governance criteria to different degrees. Adams and Abhayawansa (2022) stress that the pandemic caused the social aspects to be under scrutiny and offered a motivation for reevaluating environmental factors. Keeping that in mind, investors should consider firms in sustainability indexes as different actors and make their investment decisions by considering the firms' different approaches toward these three pillars.

Borsa Istanbul (BIST) is an emerging equity market with an expanding interest in sustainability practices and one of the founding markets that signed the United Nations Sustainable Stock Exchanges Initiative in 2012. It recommends many directives to listed companies related to sustainability performance and reporting (Ararat & Süel, 2014). The BIST Sustainability Index was launched in 2014 to lead firms to reflect on more significant concerns regarding sustainability practices, disclosure, and reporting. Moreover, the index promotes ESG practices by highlighting firms in the index to stimulate responsible investment by international and domestic investors.

Thus, this paper investigates the role of clustering firms based on their ESG scores and offering prescriptive lessons by comparatively analyzing the results of these clusters. The cluster analysis helps us in determining the natural grouping of sustainable firms (Jain, 2010). The paper considers a k-means++ algorithm in the clustering to group companies in the BIST Sustainability Index that have similar characteristics based on their ESG scores. This paper is a pioneering study in the use of a clustering approach in evaluating ESG investment. Vilas et al. (2022) employ clustering analysis to assess the inclusion and exclusion criteria used by sustainability indexes and related them to criteria considered in conventional indexes. Their results, unlike ours, show that firms with similar ESG ratings might have different characteristics regarding ESG approaches. We contribute to the literature by demonstrating the significance of forming clusters in relating sustainability practices to performance characteristics.

The structure of the paper is as follows. Section 2 provides a brief review of the literature, Section 3 presents data and methodology, Section 4 gives the results of our analysis, and Section 5 concludes with discussion and implications.

2. Literature review

Given the increased interest in sustainability issues, many papers investigate various aspects of ESG investment. Much of the literature focuses on the impact of ESG practices on their financial performance. Among the three pillars, the environmental aspect attracted the most attention. This environmental-financial performance relationship stems from the stakeholder theory and natural resource–based (NRB) perspective, which define how these practices affect the financial outcomes. The stakeholder theory suggests that dealing with the demands of several stakeholders makes firms more efficient, and environmental practices lead to better performance (Freeman & Evan, 1990). The NRB perspective conditions the firm's success on its relationship to the natural environment (Hart, 1995). Manrique and Martí-Ballester (2017) focus on a large sample of companies from developed and developing countries. They adopted various financial performance measures, such as the return on assets (ROA), the capital intensity ratio, Tobin's Q, and corporate environmental performance measures that integrate environmental approaches including emissions and resource reduction and product innovation. Their results show that environmental practices by firms have significantly positive impacts on financial performance, with a more substantial effect for firms in developing countries than in developed countries. Gupta (2018) suggests that this improved performance results from the implied cost of equity reduction due to advances in environmentally friendly approaches. Saygili et al. (2022) examine the Borsa Istanbul (BIST) Corporate Governance index and conclude that environmental disclosures reduce corporate financial performance, measured as ROA and Tobin's Q.

Several studies emphasize the role of the social pillar on performance, focusing on activities such as customer and

employee satisfaction and workplace conditions. Phan et al. (2020) note that sustainable development practices have a positive effect on financial performance directly and indirectly via customer loyalty, employee satisfaction, and corporate reputation. Edmans (2011) investigates the value-weighted portfolio of the "100 Best Companies to Work for in America" in 1984–2009 and confirms the impact of employee satisfaction on long-term stock returns. He further proves that investors prefer companies that positively affect stakeholders other than shareholders.

Studies investigating the governance pillar and firm performance relationship are numerous. Because this pillar considers the factors investigated under corporate governance, many papers examine the impact of board characteristics, control and monitoring activities, and protection of shareholder rights on performance. Following the seminal reports by Gompers et al. (2003), a mostly positive impact of these factors on performance is found in the literature. Bhagat and Bolton (2008) observe a positive impact of issues such as board members' stock ownership and CEO-chair separation on operating performance but an insignificant effect on future stock market performance. Conversely, Brown and Caylor (2006) remark that firms with higher governance scores are more profitable and more valuable and offer greater payment for their shareholders. More recently, Ciftci et al. (2019) study the Turkish context and suggested that concentrated ownership of the board size contributes to the firm performance positively. Foreign ownership is associated with better performance, whereas cross-ownership inversely affects accounting performance, and having a higher proportion of family members on the board has insignificant influence. Mertzanis et al. (2019) conclude that for a sample of 225 companies in the Middle East and North African region, board size and insider and institutional ownership are robust indicators of firm performance. Khatib and Nour (2021) investigate Malaysian firms in 2019–2020 and note that board size does not matter, whereas board diversity significantly affects firm performance during the period of uncertainty due to COVID-19. They also find a negative impact of board and audit committee meetings on performance.

Some research considers the performance of ESG investment and compares it with that of traditional financial assets. Joliet and Titova (2018) state that the higher performance of the ESG portfolios is due to their higher market level and accounting performance. Azmi et al. (2019) compare the performance of Islamic, sustainability, and Islamic sustainability equity indexes to the global equity market benchmark. They reveal that these sustainability indexes have competitive risk-return profiles, predominantly during economic expansions and bullish periods. Chiappini et al. (2021) observe that the impact of COVID-19 lockdowns on the ESG indexes was negatively affected by lockdown orders but find no statistically significant abnormal returns compared with traditional indexes. Jain et al. (2019) investigate many developed and developing markets and conclude that the sustainable and conventional indexes have become integrated. The information flow between two indexes makes them substitutes, with no statistically significant difference in performance. Santis et al. (2016) research the São Paulo Stock Exchange and corporate sustainability indexes. They find that several characteristics, such as sectoral classification rather than sustainability practices, dominate firms' financial performance. Other studies report a negative impact of ESG practices on performance (Lesser et al., 2016; Trinks & Scholtens, 2017).

Relatively few papers focus on the Turkish market. Öcal and Kamil (2021) compare the BIST All and BIST Sustainability Index to the German, French, and Indonesian sustainability indexes. The results indicate that the BIST Sustainability Index, unlike other ESG indexes, is affected by shocks more than BIST All is. Yilmaz et al. (2020) focus on membership in the BIST Sustainability Index and its relationship to market-specific performance in the period 2014–2017. Their results suggest that inclusion in a sustainability index reduces risk in terms of declines in share prices, and a positive relationship is found between inclusion and the level of ownership. Zehir and Aybars (2020) analyze the performance of ESG portfolios in Europe and Turkey and report no statistically significant performance by ESG investments. Kartal et al. (2022) discuss ways to increase the corporate governance ratings of companies listed on the BIST and stress the necessity of using advanced methods to determine the best methods for boosting ratings.

The underlying reasons for contradictory results in the literature include the methodology applied (Revelli & Viviani, 2015), the context of the sample, and, predominantly, the companies' inclusion in sustainability indexes (Arribas et al., 2021). Most papers directly relate the ESG score to a measurement of corporate performance, overlooking the different aspects of ESG. Many papers suggest that a firm in an ESG index is considered sustainable, but if the screening of the indexes for sustainability criterion is inadequate, the reported results investigating sustainability indices turn out to be inconclusive (Charlo et al., 2017). Thus, evaluating the firms' approaches to the ESG pillars is important for arriving at robust estimations. Among the many papers on ESG, Vilas et al. (2022) alone consider cluster analysis. But, unlike this study, they adopt cluster methodology to satisfy inclusion and exclusion criteria employed in sustainability indexes. This paper contributes to the literature by highlighting the differences in the practices adopted by firms in sustainability indexes.

3. Data and research methodology

3.1. Data source and description

In this paper, we perform clustering on ESG scores of the companies listed in the BIST sustainability index. The ESG company scores come from Refinitiv: a London Stock Exchange Group (LSEG) Business (Refinitiv, 2022). Refinitiv ESG scores are designed for transparently and objectively measuring a company's relative ESG performance based on publicly available and auditable data. Refinitiv ESG is a primary ESG rating provider, which has one of the most comprehensive databases in the industry (Refinitiv, 2022).

The Refinitiv ESG score includes ten components of the three pillars of sustainability. The environmental (E) pillar includes

components for "emissions," "resource use," and "innovation." The social (S) pillar, comprises components for "community," "human rights," "product responsibility," and "workforce." The corporate governance (G) pillar has components for "shareholders," "CSR strategy, and "management." The total ESG score is made up of all three pillars, whose respective weights are 34 percent (E), 35.5 percent (S), and 30.5 percent (G). Refinitiv draws on various sources to obtain ESG scores. The sources include company annual reports and websites, stock exchange filings, nongovernmental organizations' websites, corporate social responsibility (CSR) reports, and new information from ESG-trained research analysts. Although most of these sources are internal to the rated companies and tend to have a positive bias toward their practices, Refinitiv uses human and systemic processes to ensure the highest data quality.

A total of 65 companies are included in the BIST sustainability index. One of these companies (Pınar Süt Mamülleri Sanayi) is eliminated from the data set because it did not have a Refinitiv ESG score. Thus, to differentiate and group the companies based on their ESG performance, the data set initially includes scores on each of the three pillars of the 64 companies in the BIST sustainability index. After grouping these companies using a kmeans++ clustering algorithm, we collected additional data on the total ESG score, ROA, and size (ln asset) of the companies to analyze whether these clusters significantly differ from one another based on these parameters. The total ESG score comes from Refinitiv's database and the financial performance parameters from Turkiye's Public Disclosure Platform for company-level data.

3.2. Clustering algorithm: Kmeans++

In machine learning, clustering is a popular analytical method for extracting valuable information from the data. Clustering is the process of grouping similar objects or data points into clusters such that objects in the same cluster are more alike than those in other clusters. Thus, clustering divides the data set into homogeneous groups based on its properties.

Clustering is also called "unsupervised classification" because, unlike classification, it uses unsupervised learning. In unsupervised learning, a sample set cannot be labeled as a training before the clustering and the algorithm is used to identify the groups. In clustering, separate data points are categorized into clusters that represent proximate combinations of data points via a distance function.

In clustering, distance plays a crucial role. Distance quantitatively explains the similarity or dissimilarity between data points or clusters. Euclidean distance is a typical metric for measuring distance. Consider the two data points $x = (x_1, x_2, \ldots, x_d)^T$ and $y = (y_1, y_2, \ldots, y_d)^T$, where d is the dimensionality of feature space; following Plotkin (2007), the Euclidean distance between x and y is calculated as:

$$d(x,y) = \left(\sum_{j=1}^{d} (x_j - y_j)^2\right)^{1/2} \tag{1}$$

Manhattan and Canberra distance are the two other metrics for measuring distance or dissimilarities in clustering. These metrics are calculated in Eqs. (2) and (3).

$$d(x,y) = \left(\sum_{j=1}^{d} |x_j - y_j|\right) \tag{2}$$

$$d(x,y) = \frac{\left(\sum_{j=1}^{d} |x_j - y_j|\right)}{(x_j + y_j)} \tag{3}$$

As an essential and widely used clustering algorithm, k-means considers a collection of observations $(x_1, x_2, \ldots, x_n)$. Each node is a d-dimensional real vector. This algorithm divides these n observations into k sets or clusters, $(k \leq n)$, $S = \{S_1, S_2, \ldots, S_k\}$, to minimize the within-cluster sum of squared distances. Thus, k-means is obtained as follows (Celebi et al., 2013):

$$\underbrace{argmin}_{S} \sum_{i=1}^{k} \sum_{x \in S_i} \|x - \mu_i\|^2 \tag{4}$$

where μ_i are the main points in S_i. In Eq. (4), argmin is the argument of the minima or the data points of the function's domain at which the function values are minimized.

The main advantages of the k-means algorithm are simplicity, rapid convergence, and good scalability (Kwedlo & Czochanski, 2019; Meguelati et al., 2019). Because of these advantages, the k-means++ algorithm is implemented in this study. However, because the initial centers of the clusters are randomly selected, the k-means algorithm is susceptible to this selection and the chosen number of groups (k) such that it may fall into the local optimum (Huang et al., 2022; Tu et al., 2015). To improve the k-means' initial center problem, Artur and Vassilvitski (2007) proposed a k-means++ algorithm, which consists of the following steps:

Step 1. Select an initial center from $X(x_1, x_2, \ldots, x_n)$ uniformly randomly among all data points.
Step 2. Calculate D(x) for any node x (the distance between x and the center selected in step 1).
Step 3. Choose another data node randomly as a new center, employing a weighted probability distribution, whereas x is chosen with a probability proportional to $D(x)^2$.
Step 4. Repeat steps 2 and 3 until k centers have been chosen and, then, finalize the initialization.
Step 5. Because the initial centers are appropriately chosen, move through, minimizing the within-cluster sum of squared distances given by Eq. (4).

The clustering analysis is performed with the Scikit-learn tool in Python software. Another question that arises in clustering analysis is how to determine the optimal number of clusters. Different techniques are documented in the literature, among which Silhouette scores are widely used to calculate the

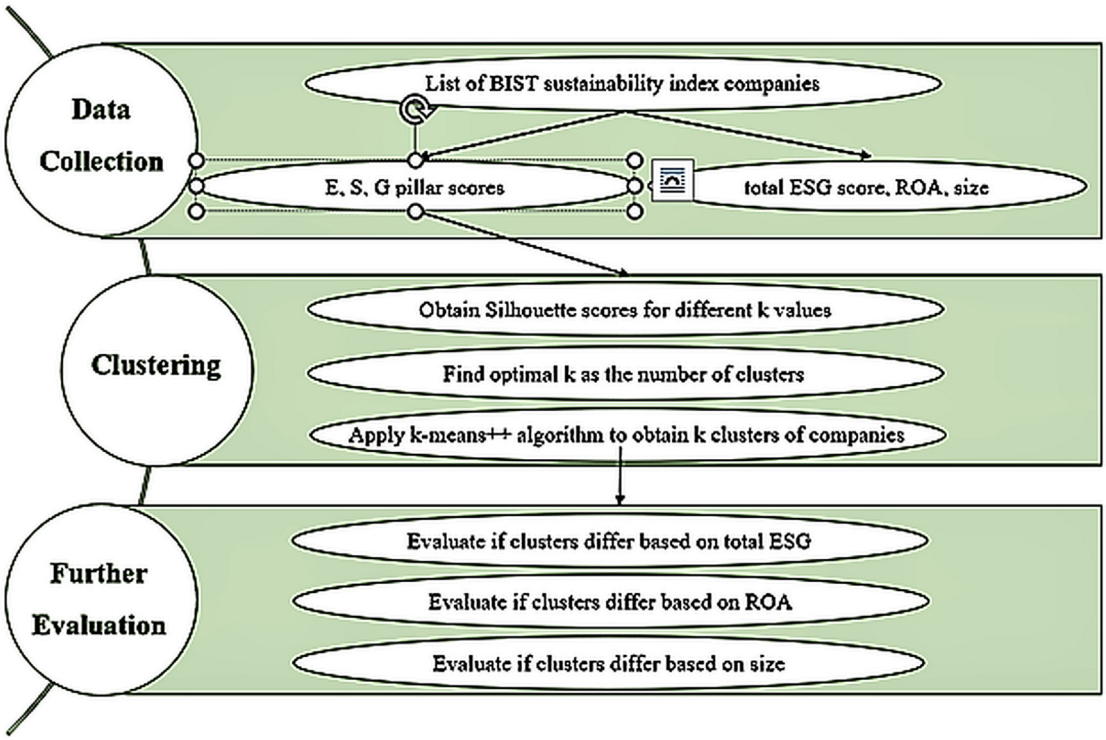

Fig. 1. Research flowchart.

optimal number of clusters (Carta et al., 2021; Johnpaul et al., 2020; Santos & Caiado, 2022). Silhouette score analysis is used to measure the distance between the resulting clusters, that is, how close each data point in one cluster is to another data point in other clusters. Silhouette scores are in the range of [−1, +1], where +1 indicates the best clustering of data points, and −1 represents data points that are not correctly clustered. Thus, positive values represent better placement of data points. Although achieving the end point of the range, +1, may not be possible in practice, the goal is to find a k value that maximizes the Silhouette score. Thus, in this data set, the optimal number of clusters is obtained using Silhouette score analysis.

In Fig. 1, the research methodology is summarized as a flow chart.

4. Empirical results

The study sample consists of E, S, and G sustainability scores for 64 companies listed on the BIST sustainability index for clustering analysis. Based on the number of companies in the sample, a k value within the range of 3–8 is chosen to obtain meaningful and practical results. Thus, the k-means++ algorithm with k values ranging from 3 to 8 is run, and the silhouette score is calculated as an output for every k value. Fig. 2 illustrates the results.

In Fig. 2, because the k value of 5 leads to the highest score, the number of clusters selected for running the k-means++ algorithm is five. This algorithm with k = 5 achieves the optimal clusters in ten iterations. The clusters obtained form a homogeneous group of companies based on E, S, and G pillar scores. Table 1 shows the centers of each cluster achieved in the final iteration.

Table 1 reports the centers of each cluster of companies for an array of ($[E_i, S_i, G_i]$) when i = 1, …,64. In other words, companies in the first cluster (k = 1) had an average E pillar score of 92.5, an S pillar score of 89.4, and a G pillar score of 70.6. The other clusters are similarly analyzed. Table 2 lists the companies and the labels assigned to clusters, enabling us to further examine the clusters obtained.

The center values (see Table 1) of the listed companies in Table 2 reveal several findings. First, although the ten companies in cluster 1 have the highest sustainability scores for the E and S pillars, they do not have the highest performance for the G pillar. Based on Table 1, the 20 companies in cluster 3 have the highest scores for the G pillar, but average scores (ranking third among the five clusters) for the E and S pillars. The firms in the first cluster are the largest and best-known companies in the Turkish context, so they engage in higher endeavors and practices than those in the other clusters to remain popular. A similar pattern is observed about the 18 companies in the fourth cluster compared to those in the first cluster: the E and S pillar scores are the second highest in this cluster, and the G pillar has nearly the lowest scores. Although the firms in the first and fourth clusters follow a similar pattern, dividing these companies into separate clusters makes sense because of the significant differences in their average scores for the three pillars. The six companies in the fifth cluster have the lowest scores in all three sustainability pillars. These firms in the fifth cluster have been targets of takeovers in the past decade or are known for having a lower level of efficiency. The

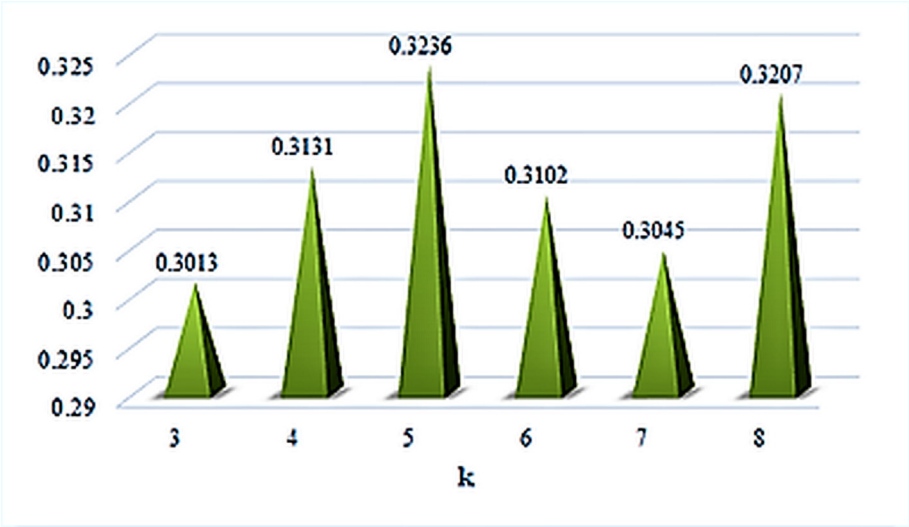

Fig. 2. Silhouette scores for k between 3 and 8.

Table 1
The centroids in the clusters for k = 5.

k	E_i	S_i	G_i
1	92.5	89.4	70.6
2	45.5	69.3	57.2
3	65.5	72.95	75.8
4	73.1	80.3	40.8
5	43.2	36.2	38.2

five clusters obtained based on these observations are labeled as in the last column of Table 2.

In a further investigation, we also study whether the clusters obtained have significant differences in parameters that are not included in the clustering analysis. The three parameters are the total ESG score of the company, ROA, and size (ln assets) to represent the overall sustainability performance, profitability, and the size of the companies, respectively, in this part of the analysis. Table 3 lists the descriptive statistics of these parameters for each of the five clusters and the significance of the differences between these clusters.

In Table 3, while the companies in the first cluster had the highest mean, minimum, and maximum total ESG scores and size (ln assets), but not ROA. However, companies in the fifth cluster had the lowest mean, minimum, and maximum total ESG scores and size (ln assets), but not ROA. The mean ESG scores of the companies grouped in the third cluster were between those in the first and fifth clusters, followed by those in the fourth and second clusters. The deviations within the clusters with respect to the total ESG scores were mainly smaller than the total variations between all the sample companies. The result for the size (ln assets) was similar, except in the first cluster. Although, in terms of size, the deviations within the second, third, fourth, and fifth clusters were smaller than the total variations, the deviation in the first cluster was greater than the total. Table 3 also shows that, although the groups significantly differed based on the mean total ESG scores and the companies' size (ln assets), no significant difference was observed among the clusters in term of ROA. Thus, the groups based on E, S, and G pillar scores of the companies listed on the BIST sustainability index displays significant differences based on total ESG scores and the size of the companies. The first cluster, which has a higher score for environmental and social practices, is the biggest in terms of assets. However, the groups are not statistically significantly

Table 2
List of companies in each cluster and labeling of clusters.

k	**n**	Company names	Cluster labels
1	10	ARCLK, ENJSA, SAHOL, KCHOL, MAVI, MIGROS, ISCTR, ULKER, YKBNK, GARAN	Companies with the highest sustainability performance for the E and S pillars and a moderate performance for the G pillar
2	10	AKSA, ANSGR, BIZIM, CIMSA, ISDMR, KARSN, SOKM, TATGD, TAVHL, TSKB	Companies with moderate sustainability performance for the G pillar and low performance for the E and S pillar
3	20	AKCNS, AKSGY, AEFES, ANHYT, AYGAZ, BIMAS, DOHOL, ENKAI, EREGLI, LOGO, PGSUS, POLHO, TKFEN, TOASO, TCELL, TUPRS, THYAO, TTRAK, VESBE, VESTL	Companies with the highest sustainability performance for the G pillar and moderate performance for the E and S pillars
4	18	AGHOL, AKBNK, AKENR, ALBRK, ASELS, BRISA, CCOLA, DOAS, FROTO, GLYHO, KERVT, KORDS, OTKAR, TTKOM, HALKB, SISE, VAKBN, ZOREN	Companies with high sustainability performance for the E and S pillars and low performance for the G pillar
5	6	AKSEN, ANELE, KRDMD, NETAS, PETKM, SKBNK	Companies with the lowest sustainability performance for the E, S, and G pillars

Table 3
Differences in total ESG scores, ROA, and size (ln asset) between clusters obtained.

parameter	Cluster id	Descriptive statistics						Significance of the difference
						95% confidence interval for mean		
		Mean	Std. dev.	Min.	Max	Lower	Upper	p-value
Total ESG score	1	84.00	4.32	76.00	92.00	80.91	87.09	0.000
	2	59.10	4.33	53.00	67.00	56.00	62.20	
	3	72.50	6.95	58.00	89.00	69.25	75.75	
	4	67.00	7.62	47.00	78.00	63.21	70.79	
	5	37.50	6.06	28.00	43.00	31.14	43.86	
	total	67.38	13.66	28.00	92.00	63.96	70.79	
ROA	1	.03	.03	−.01	.10	.01	.06	0.665
	2	.07	.07	.01	.20	.03	.12	
	3	.08	.06	−.04	.22	.05	.10	
	4	.06	.10	−.19	.24	.09	.11	
	5	.05	.14	−.18	.18	−.11	.19	
	total	.06	.09	−.19	.24	.0406	.09	
Size (ln asset)	1	18.77	2.10	15.37	20.84	17.27	20.27	0.005
	2	16.35	1.39	14.31	18.27	15.21	17.21	
	3	17.07	1.40	14.41	19.68	16.41	17.73	
	4	17.64	1.66	15.23	20.75	16.82	18.47	
	5	16.21	1.15	14.00	17.81	15.14	17.56	
	total	17.29	1.73	14.00	20.84	16.86	17.73	

different based on the overall profitability of the companies. But we can conclude that the firms with higher environmental and social scores have the lowest profitability. The firms that were the most profitable have lower E and S pillar scores and but higher scores for governance.

5. Conclusion

In this paper, we examine firms listed on the BIST Sustainability Index with a clustering approach to clarify the firms' different ESG practices. To do so, we use centroids because they offer helpful information for investors when the sustainability performances of firms is dynamic updated. When a firm's performance is updated, clustering from the beginning is not required. Instead, the new company or new performance can be placed into a cluster closer to its centroid.

The centroids can be interpreted in various ways. Although all 64 sample companies are listed on the BIST sustainability index, most of them require improvement in at least one pillar of sustainability performance. Our results demonstrate that even companies with the best business performance do not always have the highest environmental, social, and governance (ESG) scores. At the same time, low-ranking firms have better performance in one of these pillars than firms with a higher ESG rank. Thus, linking a firm with a higher ESG rank to a specific ESG pillar can produce biased outcomes. Our findings complement those by Vilas et al. (2022).

The results also suggest that firms that have higher E and S scores (cluster 1) turn out to be bigger in size. Thus larger firms are more likely to engage in environmental and social practices, so they try to have a greater impact. These companies' ROA, though insignificant, is lower than that of the other firms in the period investigated. Although cluster analysis is not concerned with the linkage, our findings are consistent with the literature that suggests an inverse relationship between corporate financial performance and the environmental and social pillars. Moreover, the group with the highest ROA (cluster 3) has the highest governance pillar scores. Thus the corporate governance practices of the firms are more likely to affect their financial performance. Our findings support the literature that suggests the environmental practices of firms have relatively little impact on performance (e.g., Saygili et al., 2022). The findings also are in line with the findings of Aydoğmuş et al. (2022) that posit the positive contribution of social and governance pillar, whereas no significant impact of environmental pillar on firm value.

The findings lead to various implications for portfolio managers and investors. While screening the ESG investments, the practices towards ESG may not be translated into profits. Thus a proper evaluation of the ESG pillars by segments is required. The investor's selection criteria might include the companies' environmental and social impact, but these activities are not necessarily related to the company's financial outcomes. However, if investors desire higher returns, they might benefit by focusing on the governance pillar, which could affect the firms' profits.

These results might hold only for this sample. Our contribution also highlights the necessity of investigating the ESG ranking of firms in clusters. Further studies could also shed more light by analyzing different firm characteristics.

Declaration of competing interest

The authors declare no conflict of interest.

Appendix.

Full company names

Cluster	Company names
1	Arcelik AS, Enerjisa Enerji AS, Hacı Ömer Sabancı Holding AS, Koc Holding AS, Mavi Giyim Sanayi ve Ticaret AS, Migros Ticaret AS, Türkiye Garanti Bankası AS, Türkiye İş Bankası AS, Ülker Bisküvi Sanayi AS, Yapı ve Kedi Bankası AS, Garanti Bankası AS
2	Aksa Akrilik Kimya Sanayi AS, Anadolu Anonim Türk Sigorta Sti, Bizim Toptan Satış Mağazaları AS, Cimsa Çimento Sanayi ve Ticaret AS, İskenderun Demir ve Çelik AS, Karsan Otomotiv Sanayii ve Ticaret AS, Sok MArketler Ticaret AS, Tat Gıda Sanayi AS, TAV HAvalimanları Holding AS, Türkiye Sinai Kalkınma Bankası AS
3	Akcansa Cimento Sanayi ve Ticaret AS, Akis Gayrimenkul Yatirim Ortakligi AS, Anadolu Efes Biracilik ve Malt Sanayii AS, Anadolu Hayat Emeklilik AS, Aygaz AS, BIM Birlesik Magazalar AS, Dogan Sirketler Grubu Holding AS, ENKA Insaat ve Sanayi AS, Eregli Demir ve Celik Fabrikalari TAS, Logo Yazilim Sanayi ve Ticaret AS, Pegasus Hava Tasimaciligi AS, Polisan Holding AS, Tekfen Holding AS, Tofas Turk Otomobil Fabrikasi AS, Turkcell Iletisim Hizmetleri AS, Turkiye Petrol Rafinerileri AS, Turk Hava Yollari AO, Turk Traktor ve Ziraat Makineleri AS, Vestel Beyaz Esya Sanayi ve Ticaret AS, Vestel Elektronik Sanayi ve Ticaret AS
4	AG Anadolu Grubu Holding AS, Akbank TAS, Akenerji Elektrik Uretim AS, Albaraka Turk Katilim Bankasi, Aselsan Elektronik Sanayi ve Ticaret AS, Brisa Bridgestone Sabanci Lastik Sanayi ve Ticaret AS, Coca-Cola Icecek AS, Dogus Otomotiv Servis ve Ticaret AS, Ford Otomotiv Sanayi AS, Global Yatirim Holding AS, Kerevitas Gida Sanayi ve Ticaret AS, Kordsa Teknik Tekstil AS, Otokar Otomotiv ve Savunma Sanayi AS, Turk Telekomunikasyon AS, Turkiye Halk Bankasi AS, Turkiye Sise ve Cam Fabrikalari AS, Turkiye Vakiflar Bankasi TAO, Zorlu Enerji Elektrik Uretim AS
5	Aksa Enerji Uretim AS, Anel Elektrik Proje Taahhut ve Ticaret AS, Kardemir Karabuk Demir Celik Sanayi ve Ticaret AS, Netas Telekomunikasyon AS, Petkim Petrokimya Holding AS, Sekerbank TAS

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